17.4 部分习题参考简答

1. (1) $\dfrac{1}{x-1}=-\dfrac{1}{2}\sum_{n=0}^{\infty}\left(\dfrac{x+1}{2}\right)^n, \quad x\in(-3,1)$；

(2) $\dfrac{x}{(x-1)(x+3)}=\dfrac{1}{4}\sum_{n=0}^{\infty}\left(\left(-\dfrac{1}{3}\right)^n-1\right)x^n, \quad x\in(-1,1).$

(3) $f(x)=\sum_{n=0}^{\infty}(-1)^n\dfrac{x^{2n+1}}{(2n+1)^2}, \quad x\in[-1,1].$

(4) $\dfrac{1}{1+x+x^2}=\dfrac{4}{3}\sum_{n=0}^{\infty}(-\dfrac{4}{3})^n\left(x+\dfrac{1}{2}\right)^{2n}, \quad x\in(-\dfrac{\sqrt{3}+1}{2},\dfrac{\sqrt{3}-1}{2}).$

2. (1) $\dfrac{\ln(1-x)}{1-x}=-\sum_{n=0}^{\infty}(\sum_{k=1}^{n}\dfrac{1}{k})x^n, \quad (|x|<1)$；

(2) $\dfrac{1}{(1-x)^2}=\sum_{n=1}^{\infty}nx^{n-1}, \quad (|x|<1).$

(3) $f(x)=\sum_{n=0}^{\infty}\dfrac{x^{4n+1}}{4n+1}, \quad (|x|<1).$

(4) $\dfrac{d}{dx}\left(\dfrac{e^x-1}{x}\right)=\sum_{n=1}^{\infty}\dfrac{nx^{n-1}}{(n+1)!}, \quad (x\in\mathbb{R}).$

(5) $\ln^2(1+x)=2\sum_{n=2}^{\infty}\dfrac{(-1)^n}{n}(1+\dfrac{1}{2}+\dfrac{1}{3}+\cdots+\dfrac{1}{n-1})x^n, \quad x\in(-1,1].$

3. (1) $1+x+\dfrac{1}{2}x^2$.　　(2) $x+x^2+\dfrac{1}{3}x^3$.

4. $f^{(n)}(0)=\begin{cases}-\dfrac{1}{2}, & n=1,\\ \dfrac{3n!}{(-2)^n}, & n\ge 2.\end{cases}$　　$g^{(n)}(0)=\dfrac{n!}{3}\left(\left(-\dfrac{1}{2}\right)^n-1\right).$

5. 提示：$\dfrac{1}{1-3x+2x^2}=\dfrac{1}{(1-2x)(1-x)}=\dfrac{2}{1-2x}-\dfrac{1}{1-x}.$